

DS 6021: Intro to Predictive Modeling

Lab 3: Feature Selection:

Multicollinearity, Categorical Predictors, Interactions, and Outliers

September 18, 2026

Due: September 20, 2026 11:59pm

Materials Required

All available in our class github repository:

- penguins.csv – Palmer penguins dataset
- bike-sharing-daily.csv – Daily bike rental counts, Washington DC, 2011–2012 (UCI Bike Sharing Dataset; the same data used in Christoph Molnar's Interpretable Machine Learning)
- In-class notebooks from 9/14 (feature_selection_pt1_categorical_vars.ipynb) and 9/16 (feat_selection_pt2_interactions_outliers.ipynb)

Activity Overview

In class this week we asked which features belong in a model, and looked specifically at multicollinearity, categorical predictors, and interaction terms. We also looked at outliers, leverage, and influence on the bike rentals data. We will continue these exercises in this lab. You will also do a little bit of reading on feature transformations, which we didn't have time to get to in class.

Throughout the penguin questions, you will first need drop rows missing any of bill_length_mm, bill_depth_mm, flipper_length_mm, or body_mass_g as we've been doing in multiple homeworks and in-class notebooks so far.

Part 1: Multicollinearity and Multiple Linear Regression (9 pts)

1. [2 pts] Last week, in Part 4b, I asked you to compare three models predicting penguin body mass and based on the R^2 , adjusted R^2 , RSE, and the F-statistic, I asked you to explain which model you would choose, and why. Given the techniques we discussed in Week 4 in class, how would you determine which model is better? Does your answer change from what you listed in last week's homework?
2. [7 pts] In class, when looking at the bike data, we showed that temp and atemp ("feels like") temp are highly correlated, but we didn't explicitly check the correlation of other variables in our model. Now switch to the bike data. For this question, we want to explicitly look at $\text{cnt} \sim \text{temp} + \text{hum} + \text{windspeed}$
 - a. [1 pt] Look at the correlation matrix for all of these variables. In addition to the .corr() function, use the sm.heatmap() function for an equivalent visual.
 - b. [2 pts] Build a linear model as we did in class predicting $\text{cnt} \sim \text{tmp} + \text{hum} + \text{windspeed}$. Calculate the VIF for each of these features. For each feature (temp, hum, windspeed) describe how it impacts the count of bikes rented. Does this match your intuition?
 - c. [2 pts] Build three models for count and use the nested F-test to compare them. Although we did not do this explicitly in class, the anova_lm function will accept three models as arguments.
 - i. $\text{cnt} \sim \text{tmp}$

- ii. $\text{cnt} \sim \text{tmp} + \text{hum}$
- iii. $\text{cnt} \sim \text{tmp} + \text{hum} + \text{windspeed}$
- d. [2 pts] Of these three models, which model would you choose and why?

Part 2: Categorical Predictors and Interaction Terms (15 pts)

For the first part of this section, we will be looking at the penguin data with sex as a categorical variable. This means you will need to modify the penguin dataset to drop the penguins where sex is missing.

1. [3 pts] Fit $\text{body_mass_g} \sim \text{flipper_length_mm} + \text{bill_depth_mm} + \text{bill_length_mm}$ with the new dataset (e.g. with additional penguins removed where sex is missing). Compare the output of this regression with the one you did previously where the penguins where sex is missing were not removed. Does removing these penguins change anything about the model?
2. [3 pts] Fit an additive model for $\text{body_mass_g} \sim \text{flipper_length_mm} + \text{C}(\text{sex})$ as well as an interactive model. Compare these two models with the model that only includes flipper_length.
 - a. Which of the three models is the best and why? How did you come to this decision?
 - b. What is the impact of sex on the prediction of body mass? How do male and female penguins differ?
 - c. Include at least one plot and explain how you used it to help you answer this question.
3. [3 pts] Fit additive and interactive models for body_mass_g using only sex and species as predictors.
 - a. Which model is better?
 - b. How would you interpret the coefficients in this case?
 - c. Include at least one plot and explain how you used it to help interpret this model.
4. [6 pts] In class, we predicted count of bike rentals with working day as a categorical predictor to identify if there was a significant interaction with temperature. (We found there was not.) Now we are going to look at casual and registered users separately.
 - a. Fit a model: $\text{casual} \sim \text{temp} + \text{C}(\text{workingday}) + \text{hum} + \text{windspeed}$. Then build a second model that includes the interaction between temp and working day. Compare these two models. Which is better?
 - b. Repeat part a) but with registered instead of casual as the thing you are predicting.
 - c. Interpret the results for a and b. What is this data telling you.
 - d. Why did I ask you to build two separate models vs adding either of these variables into our original model?

Part 3: Outliers, Leverage, and Influence (12 pts)

1. [8 pts] In part 2 above, we removed 9 penguins from the data set in order to look at sex as a predictor variable. Using the original model predicting $\text{body_mass_g} \sim \text{flipper_length_mm} + \text{bill_depth_mm} + \text{bill_length_mm}$ (n=342) do the following:
 - a. Build a DataFrame, similar to the one we did in class, that contains the following for each penguin: actual body mass, predicted body mass, studentized residuals, leverage, and Cook's distance.

- b. Report the cutoffs we used in class and identify the number of penguins that exceed each cutoff, if any. Pay particular attention to the 9 penguins that are missing the sex variable. Do any of them exceed any of these cutoffs?
 - c. Reflect on your answer in part 2. Does the information you have on studentized residuals, leverage, and Cook's distance support or contradict what you concluded earlier? Would you change anything about your answer?
 - d. Create a plot like the one we did in class where you can look at leverage, studentized residuals, Cook's D. Share your observations.
2. **[4 pts]** In class, we built a model and looked at outliers, high leverage, and influential datapoints for the bike rental data set. We specifically identified 10/29/2012 as in the window for Hurricane Sandy. Given the model we have built, what happens to the coefficients if we remove this particular data point? What happens if we remove the data point from 2012-03-17, which meets the threshold for all three metrics? Would you remove either of these data points from your model, and if you did, how would you report on it to a stakeholder?

Part 5: Feature Transformations (5 pts)

We ran out of time for this in class, so this part of the lab will explore data transformations in more detail. You have already done one transformation in this course: the log-log model for the planets at the end of Lab 2. We also explored centering of the data in class when we looked at interaction terms.

For this portion of the lab, please do not use any Generative AI.

Please read the following references:

- <https://library.virginia.edu/data/articles/interpreting-log-transformations-in-a-linear-model>
- <https://machinelearningmastery.com/feature-scaling-in-practice-what-works-and-what-doesnt/>
- https://scikit-learn.org/stable/data_transforms.html
 - Specifically, please focus on sections 8.3.1 and 8.3.3, you are not required to read all of this!

Please write a short paragraph (around 5 sentences) reflecting on these readings and relate to at least one topic we have discussed in class. Please also write one question about feature transformations that you have after reading this material. (This is a very open ended portion of the lab!)

Deliverables

For each part, please include the following:

- All code used to complete the activities. This can be turned in as a single Jupyter notebook, multiple Jupyter notebooks, or one or more .py files. Up to you!
- For open ended questions, please include a few sentences or a paragraph answering the question. This text can be included in your Jupyter notebook as a markdown cell or a separate document.
- Per our policy on Generative AI detailed in the syllabus, if you use it to generate code, you need to flag how and where you used it, and how you ascertained the correctness of what you did. If you did not use Generative AI for any part of the assignment, please state that explicitly.